

**Here in this part of course,** you'll learn to build fully functional web applications using Bubble.io, a no-code/low-code platform that lets you visually design interfaces, define workflows, and manage data without writing code. You will explore essential concepts including pages, elements, data types, and workflows, along with the fundamentals of structuring databases and designing responsive layouts using Bubble's flexbox-based engine.

**About Instructor:**

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**About Institute:**

NexSkill is Pakistan's leading IT training institute, providing the best IT training in Pakistan. NexSkill offers a variety of professional learning platforms to its students.

<https://www.nexskill.com/>

- ❑ Bubble.io is a no-code/low-code platform\*\* that allows users to build fully functional web applications without writing traditional code, using a visual drag-and-drop editor and workflow logic.
- ❑ It provides tools for UI design, database creation, workflows, APIs, and deployment—all within one unified platform.

## **No-Code / Low-Code Fundamentals & Use-Cases\*\***

No-code/low-code platforms \*\*democratize software development\*\*, enabling entrepreneurs, designers, and non-technical users to create digital products without needing programming skills.

These platforms are ideal for:

- ✓ Rapid prototyping and MVP development
- ✓ Launching startups or testing business ideas quickly
- ✓ Building internal tools and business automation
- ✓ Creating scalable web apps without a full engineering team



- ❑ Bubble supports building complex and scalable web applications\*\*, including social networks, marketplaces, directories, productivity tools, and more.  
<https://medium.com/@thomasczerny/bubble-io-an-invaluable-no-code-low-code-skill-you-need-to-know-1abfc09be117>
- ❑ Training examples from Bubble courses include:
  - ❑ A Tinder-style matching app built using Bubble layouts, workflows, and user data models.  
<https://www.udemy.com/course/build-fully-functional-apps-without-coding-using-bubble/>
  - ❑ A Twitter-like social network demonstrating database structuring, user profiles, timelines, and interactions. [<https://www.keyskillset.com/nocodify>]
- ❑ Bubble Academy highlights that creators can build native mobile apps, AI-powered apps, and production-ready systems, showcasing the platform's versatility. <https://bubble.io/academy>





- ❑ **Design Tab** — Used to build and customize the visual layout of your app using Bubble's drag-and-drop editor. This is where you add UI elements, arrange containers, and structure responsive layouts.
- ❑ **Workflow Tab** — Defines app logic through event-driven actions such as button clicks, data changes, and navigation. It controls how the app behaves in response to user interactions.
- ❑ **Data Tab** — Allows you to manage your database, configure data types, set fields, create privacy rules, and view live app data. Essential for handling dynamic content.
- ❑ **Styles Tab** — Provides centralized styling for text, buttons, and visual elements, ensuring consistent design by applying reusable style definitions across your app.  
<https://www.youtube.com/watch?v=N2Lcke85fHM>
- ❑ **Settings** — Includes application configuration such as SEO, domain setup, email settings, API keys, version control, and general app-level options.  
<https://www.udemy.com/course/build-fully-functional-apps-without-coding-using-bubble/>
- ❑ **Logs** — Displays workflow execution logs, performance metrics, and debugging information to help diagnose issues and optimize app behavior.  
<https://www.udemy.com/course/build-fully-functional-apps-without-coding-using-bubble/>



- ❑ **Plugins extend Bubble's functionality:**, allowing creators to integrate external services, add new UI elements, automate actions, and enhance app capabilities without custom coding. <https://www.udemy.com/course/build-fully-functional-apps-without-coding-using-bubble/>
- ❑ **Connecting your Bubble app to thousands of external services** (e.g., APIs, automation tools) is made possible through plugin integrations, enabling features like email delivery, OAuth, file handling, and third-party automation. [<https://www.youtube.com/playlist?list=PLd6fHsjyvH7z4Z8yCsp8Uu6MFqPI95KrW> ]
- ❑ **Bubble plugins** cover a wide range of functions, including:
  - ✓ Security & authentication tools (e.g., OAuth providers)
  - ✓ Workflow enhancers, such as backend workflow tools and custom event plugins
  - ✓ Email & communication services, such as Postmark for transactional messages
  - ✓ Automation & tracking utilities, like LogSnag for event logging or BannerBear for media creation<https://www.youtube.com/playlist?list=PLd6fHsjyvH7z4Z8yCsp8Uu6MFqPI95KrW>
- ❑ **Bubble App Marketplace** provides both free and paid plugins to help integrate APIs, expand database features, improve UI components, or streamline workflows—making Bubble a highly flexible no-code platform. <https://bubble.io/academy>



## ❑ Pages

Bubble apps are built on individual pages that act as screens for users. Pages can display static or dynamic content and are essential for structuring navigation within the application. Dynamic pages can change content based on user inputs or URL parameters.

## ❑ Elements

Elements are the UI building blocks of a Bubble app—such as text, buttons, images, groups, input fields, and reusable components. These elements are assembled visually through the design editor and customized to create user interfaces.

## ❑ Data Types

Data types define the structure of the app's database. Each data type (e.g., User, Post, Product) contains fields that store information. They form the backbone of dynamic application content and logic.

## ❑ Workflows

Workflows determine how the application behaves. They define event-driven logic—such as what happens when a user clicks a button, submits a form, or loads a page. Workflows power interactivity and automation throughout the app.

<https://www.youtube.com/watch?v=XAoCmqtvzhA>





## ❑ **Creating and Structuring Databases**

Bubble allows you to create custom data types and fields to structure your app's database. Proper database design is foundational for storing, retrieving, and managing dynamic content within the application. [

<https://www.youtube.com/watch?v=XAoCmqtvzhA> ]

## ❑ **Data Relationships, Option Sets, and Privacy**

Bubble supports relational data models, enabling you to link data types together (e.g., Users → Posts). Option sets provide fixed-value lists for cleaner database logic, and Bubble includes privacy rules that control which users can see or modify specific data.

## ❑ **Working with Database Triggers and Backend Workflows**

Bubble includes backend workflows and database triggers that automate operations in the background—such as running scheduled tasks, detecting data changes, and automating system logic without user interaction.

<https://www.youtube.com/playlist?list=PLd6fHsjyvH7z4Z8yCsp8Uu6MFqPI95KrW>



## Responsive design using the flexbox responsive engine

Bubble's updated responsive engine (based on flexbox principles) enables you to build layouts that adapt smoothly to different screen sizes. It allows for flexible alignment, spacing, and stacking behaviors, making it easier to create modern, mobile-friendly interfaces.

<https://buildingwithbubble.com/complete-bubble-crash-course-beginners/>

## Styles and theming

Bubble provides a centralized \*Styles\* system where you can define reusable visual formats for elements (e.g., text, buttons, inputs). This enables consistent theming across pages and speeds up UI development, while allowing global updates from a single place.

<https://www.udemy.com/course/build-fully-functional-apps-without-coding-using-bubble/>





## Dynamic pages

Bubble allows you to create \*dynamic pages\* whose content changes based on user input or URL parameters—ideal for user profiles, product pages, or any data-driven content. [ <https://www.youtube.com/watch?v=XAoCmqtvzhA> ]

## Repeating groups

Repeating groups enable you to display lists of items dynamically (e.g., posts, products, messages). They automatically pull data from the database and update as new items are added or changed.

<https://www.youtube.com/watch?v=XAoCmqtvzhA>

## Filtering data, search, and sorting

Bubble's data tools allow you to filter, search, and sort database entries directly within repeating groups and dynamic content areas, helping users find relevant information efficiently. (Filtering and display functionality are part of Bubble's dynamic content and repeating group features.)

<https://www.youtube.com/watch?v=XAoCmqtvzhA>



### ❑ **Connecting external APIs**

Bubble enables integration with thousands of external services and APIs, allowing apps to connect to automation platforms, external databases, and SaaS tools. This is accomplished through Bubble's API connector and plugin ecosystem.

### ❑ **Using plugins for automation, email, file handling**

Plugins greatly expand app capabilities—supporting automated workflows, secure file processing, and advanced user interactions. Bubble's plugin library includes tools for connecting workflows, sending emails, managing uploads, and other automation tasks.

### ❑ **Postmark (Transactional Email)**

The Postmark plugin allows apps to send reliable transactional emails, notifications, and system messages directly through workflows—useful for sign-ups, password resets, and alerts.

### ❑ **LogSnag (Event Logging & Notifications)**

Plugins like LogSnag help track real-time user actions, log events, and send notifications—useful for monitoring user activity and maintaining analytics within Bubble apps.

<https://www.youtube.com/playlist?list=PLd6fHsjyvH7z4Z8yCsp8Uu6MFqPI95KrW>





# Thanks for your time

**Bubble.io** - No-Code Web Site, CMS , App , Business solution – Platform

